Making Connections: Can external knowledge help models solve NYT Connections?

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Abstract

NYT Connections is a word-grouping puzzle in which players must partition 16 words into four hidden categories. The task is challenging for artificial intelligence because successful solutions often require lexical knowledge, cultural references, and semantic reasoning beyond simple word similarity. In this work, we investigate whether external knowledge can improve performance on NYT Connections puzzles. We compare three embedding-based approaches: a baseline Sentence-BERT clustering model, a WordNet-enhanced model, and a multi-enhanced knowledge-augmented model that combines WordNet and Wikipedia context through embedding fusion. Experiments on over 900 archived NYT Connections puzzles show that external knowledge consistently improves clustering performance. The best-performing model increased Adjusted Rand Index (ARI) from 0.198 to 0.251 and pairwise accuracy from 0.710 to 0.737, significantly outperforming both the baseline and WordNet-only models. These results suggest that many failures on semantic grouping tasks stem from missing contextual knowledge rather than weak semantic reasoning, and that lightweight knowledge augmentation can substantially improve performance without additional model training.

1. Introduction

1.1. Context & Background

The New York Times (NYT) Connections puzzle has emerged as a globally popular daily word game. The premise is deceptively simple: players are presented with a grid of 16 words and must partition them into four distinct groups of four, where each group shares a common, unifying semantic thread. The puzzle is made complex with its intentional multi-layered ambiguity. Words are explicitly selected because they possess multiple meanings, often belonging to overlapping categories, or rely on sophisticated lexical properties such as homophones, anagrams, slang, and niche pop-culture references [Figure 1].

From a computational linguistics perspective, Connections is a departure from traditional natural language processing (NLP) tasks. Standard sequence-to-sequence models or classification frameworks operate on dense textual contexts where syntax, word order, and surrounding sentence structures provide rich semantic clues. In contrast, Connections presents words in complete isolation, so to solve this, an agent can't rely on localized syntactic parsing. Instead, it must perform high-level relational reasoning across the grid, evaluating competing hypotheses and resolving global interference where a single word could logically fit into multiple categories.

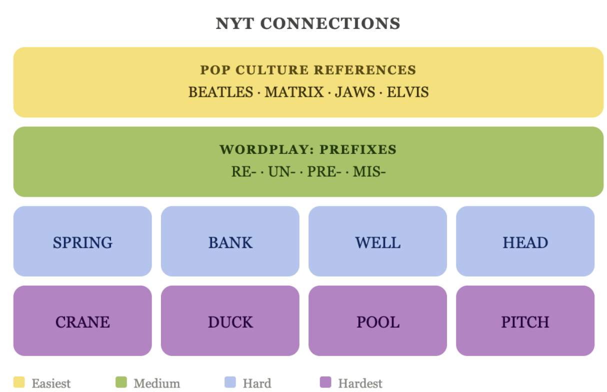

Figure 1. An example puzzle grid demonstrating the lexical ambiguity and categorical overlap (e.g., pop culture vs. wordplay prefixes) inherent to the Connections task.

1.2. Motivation

While modern language models have achieved impressive performance on a wide range of language understanding tasks, prior work has shown that they often struggle with NYT Connections puzzles [1, 2]. This observation raises an important question: do models fail because they cannot identify the underlying semantic relationships between words, or because they lack the contextual knowledge required to recognize those relationships in the first place?

For example, a model may fail to group a set of words because it lacks knowledge of a specific cultural reference, historical figure, or brand name. In this case, the failure stems from missing information rather than an inability to reason about semantic similarity. Conversely, if a model possesses relevant lexical and contextual knowledge but still fails to recover the correct categories, then the limitation may lie in how semantic relationships are represented and clustered.

Understanding this distinction is important for developing systems that can effectively combine learned semantic representations with external knowledge sources. By investigating how structured knowledge from resources such as WordNet and Wikipedia affects performance on NYT Connections, we can better understand the role that contextual knowledge plays in semantic grouping tasks and determine whether knowledge augmentation can help bridge existing performance gaps.

1.3. Research Questions

To explore these challenges, this project investigates the following primary research question:

How does access to domain-specific external knowledge impact a model's ability to correctly group words in NYT Connections puzzles?

We also break this question down into the following secondary sub-question:

Which model architectures best capture contextual similarity and resolve semantic overlap when executing multi-class word grouping tasks?

1.4. Related Works

NYT-CONNECTIONS: A Deceptively Simple Text Classification Task that Stumps System-1 Thinkers [1] views NYT Connections as a benchmark for evaluating reasoning and semantic understanding in language models. The paper states that the "NYT Connections uniquely combines linguistic isolation, resistance to intuitive shortcuts, and regular updates to mitigate data leakage, offering a novel tool for assessing LLM reasoning capabilities." Therefore, Connections puzzles are resistant to shortcut learning, because they require deeper contextual and semantic reasoning rather than simple pattern recognition. This work is relevant to our project, because it narrows our focus to whether model failures arise from weak reasoning capabilities or missing contextual knowledge.

Learning Semantic Complexities of NYT Connections [2] explores different NLP methods that can be used for solving Connections puzzles, such as clustering algorithms, embedding-based similarity approaches, and fine-tuned language models. In this paper, we view Connections as a semantic grouping problem and evaluate how different architectures capture linguistic and contextual relationships between words. The work is relevant to our project, because it provides baseline methodologies and evaluation strategies that we can compare against when testing embedding-based and context-enhanced models.

A Semantic Approach for Text Clustering Using WordNet and Lexical Chains [3] explores improving clustering performance by incorporating semantic relationships from WordNet rather than relying only on statistical similarity measures. This paper shows how semantic knowledge sources can improve clustering quality and better capture conceptual meaning between words. This work is relevant to our project, because we also want to investigate whether adding external semantic knowledge from WordNet and other resources will improve a model's ability to correctly group words in Connections.

2. Methods

2.1. Dataset

We utilized a Kaggle dataset [4] containing over 900 archived NYT Connections puzzles collected since the game's release on June 12, 2023. Each puzzle consists of 16 words partitioned into four ground-truth categories.

The raw dataset is organized at the word level, with each row containing a puzzle identifier, word, category label, and difficulty tier. Because our models operate at the puzzle level, we grouped words by puzzle identifier and reconstructed each puzzle into its complete 16-word format along with its corresponding category assignments.

We applied a validation procedure to ensure that each puzzle contained exactly 16 unique words, four categories, and four words per category. Puzzles that failed these checks were removed from the dataset.

After preprocessing, the dataset was randomly partitioned into training, validation, and test sets using a 70/15/15 split. This resulted in 644 puzzles in the training set and 138 puzzles each in the validation and test sets. The validation set was used for hyperparameter selection and model tuning, while the held-out test set was reserved exclusively for final evaluation.

2.2. Preprocessing

To clean the data, we first standardized each word by converting it to lowercase for consistency, and removing leading or trailing whitespace for. We then validated each puzzle to ensure that it followed the NYT Connections format: exactly 16 unique words, exactly four groups, and four words per group. Any invalid puzzles that broke this structure were removed. Next, we mapped each qualitative group name to a numerical label from 0 to 3, to represent the four color mappings of the NYT Connections' groupings, allowing clustering outputs to be compared against ground-truth labels. Finally, we restructured the dataset so that each row represented a complete puzzle rather than a single word. Each puzzle-level row stored the game ID, list of words, baseline input texts, category mapping, true labels, and group difficulty levels. This pipeline is illustrated in Figure 2 below.

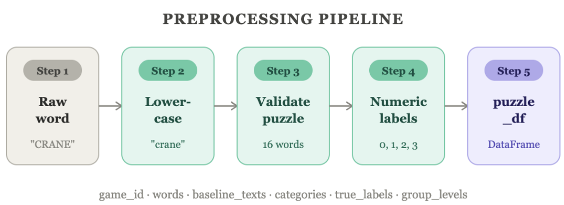

Figure 2. Flow diagram of the data preprocessing pipeline, outlining the transition from raw text to structured puzzle-level inputs.

2.3. Baseline Sentence-BERT Model

Our baseline model relies solely on pretrained Sentence-BERT embeddings and evaluates how well semantic similarity alone can solve NYT Connections without any external knowledge augmentation [2]. For each puzzle, the 16 raw puzzle words were embedded using the Sentence-BERT model all-MiniLM-L6-v2 [Figure 3]. This model was selected because it is specifically designed for semantic similarity tasks, producing embeddings in which semantically related inputs are positioned close together in vector space. Additionally, all-MiniLM-L6-v2 provides strong performance while remaining computationally efficient, generating compact 384-dimensional embeddings that enabled extensive experimentation across multiple model variants and hyperparameter settings. We used this same embedding model across all experimental conditions to ensure that any observed performance differences could be attributed to the addition of external knowledge rather than changes in the underlying encoder.

Given a puzzle containing 16 words, Sentence-BERT generated one embedding vector for each word. Embeddings were L2-normalized during encoding so that cosine similarity could be used as a consistent measure of semantic relatedness between words. The resulting embedding matrix therefore contained 16 normalized vector representations, one for each puzzle word.

Rather than treating Connections as a classification problem, we formulated it as a clustering task. Since each NYT Connections puzzle contains exactly four hidden categories, we applied Agglomerative Clustering with the number of clusters fixed at four. Agglomerative clustering begins by assigning each word to its own cluster and then iteratively merges the most similar clusters until only four clusters remain. We used cosine distance as the clustering metric because it aligns naturally with Sentence-BERT embeddings and captures semantic similarity between words.

To determine the best clustering configuration, we evaluated three linkage strategies—single, average, and complete linkage—on the validation set. The linkage method controls how distances between clusters are computed during the merging process. The configuration producing the highest validation-set Adjusted Rand Index (ARI) was selected and subsequently used for all train, validation, and test evaluations.

Model performance was evaluated using four complementary metrics. Exact Match Accuracy measures whether all four predicted groups exactly match the ground-truth puzzle solution. Pairwise Accuracy evaluates whether the model correctly predicts whether every pair of words belongs to the same group, yielding 120 pairwise comparisons per puzzle.

Adjusted Rand Index (ARI) measures clustering similarity while correcting for chance agreement [1] Normalized Mutual Information (NMI) measures how much information is shared between predicted and true cluster assignments. Together, these metrics provide both strict and partial measures of puzzle-solving performance.

This baseline serves as the control condition for determining whether external knowledge improves semantic grouping performance on NYT Connections puzzles.

2.4. WordNet-Enhanced Model

The WordNet-enhanced model investigates whether explicit lexical knowledge improves semantic grouping performance. While the baseline model relies solely on the semantic information already encoded within pretrained Sentence-BERT embeddings, this model augments each word with definitions obtained from WordNet before embedding generation [3].

For every puzzle word, we queried WordNet and retrieved up to three synsets (definitions) corresponding to possible meanings of that word. We then extracted the definition associated with each synset and concatenated these definitions into a single context string. For example, an ambiguous word such as crane may correspond to both a bird and a construction machine; by appending multiple definitions, the model gains access to several possible interpretations rather than relying solely on the surface form of the word. If no WordNet synsets were available for a word, the original word was retained without modification.

The resulting context-enhanced representation consisted of the original word followed by the retrieved WordNet definitions. These augmented texts were then encoded using the same Sentence-BERT model (all-MiniLM-L6-v2) used in the baseline condition [Figure 3]. Embeddings were normalized during encoding and subsequently clustered using Agglomerative Clustering with the number of clusters fixed at four.

To ensure a fair comparison with the baseline model, all components of the pipeline—including the embedding model, clustering algorithm, and evaluation metrics—were kept identical. The only difference between the two models was the addition of WordNet-derived lexical context. As with the baseline model, we evaluated single, average, and complete linkage strategies on the validation set and selected the configuration that achieved the highest validation-set ARI before performing final train, validation, and test evaluations.

By introducing explicit lexical definitions while holding all other components constant, this model isolates the contribution of WordNet knowledge and allows us to determine whether providing additional semantic context improves a model's ability to recover the hidden categories within NYT Connections puzzles.

2.5. Multi-Enhanced Model

The multi-enhanced model combines lexical and encyclopedic context by fusing three separate representations of each puzzle word: the raw word, the WordNet-enhanced text, and the Wikipedia-enhanced text. This model was designed to test whether combining multiple forms of external knowledge produces stronger semantic groupings than either the baseline or WordNet-only model [1].

For each word, we first generated the same raw Sentence-BERT embedding used in the baseline model. We then generated a WordNet-enhanced embedding by encoding the word concatenated with up to three WordNet definitions. Finally, we generated a Wikipedia-enhanced embedding by encoding the word together with a short Wikipedia summary when available [Figure 3]. Wikipedia summaries were automatically retrieved for every unique puzzle word and stored in a local cache to avoid repeatedly querying Wikipedia during experimentation. Caching substantially reduced computation time and enabled efficient evaluation of multiple model configurations.

To investigate the effect of context length, Wikipedia summaries were initially cached at 300 characters and subsequently truncated to create 100-character, 200-character, and 300-character variants. These variants were treated as a hyperparameter and evaluated independently during model selection.

The three embeddings were combined using a weighted fusion strategy:

$$E_{\text{fused}} = \alpha E_{\text{word}} + \beta E_{\text{WordNet}} + \gamma E_{\text{Wikipedia}}$$

where E_{word} is the raw word embedding, E_{WordNet} is the WordNet-enhanced embedding, and $E_{\text{Wikipedia}}$ is the Wikipedia-enhanced embedding. The fusion weights α, β, and γ were treated as hyperparameters and tuned on the validation set. After fusion, the resulting vectors were ℓ_2-normalized to preserve cosine similarity behavior before clustering. The fused embeddings were then passed into the same agglomerative clustering pipeline used by the baseline and WordNet-only models, with the number of clusters fixed at four. Multiple linkage strategies were evaluated during model selection. By keeping the embedding model, clustering algorithm, and evaluation metrics consistent across all models, this architecture isolates the effect of combining lexical and encyclopedic external knowledge on NYT Connections grouping performance.

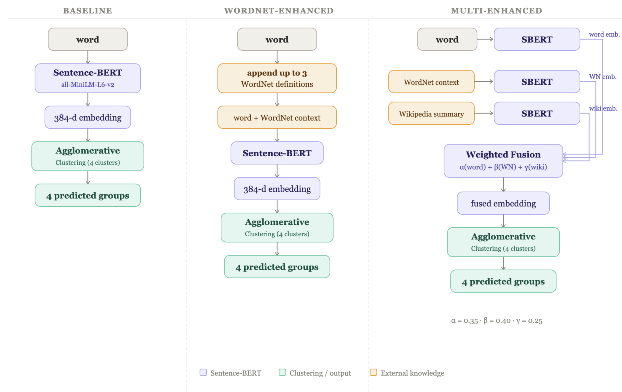

Figure 3. Overview of the baseline, WordNet-enhanced, and multi-enhanced architectures. All three models use Sentence-BERT embeddings and agglomerative clustering, while the latter two incorporate external knowledge through WordNet definitions and Wikipedia summaries. The multi-enhanced model fuses semantic, lexical, and encyclopedic representations before clustering into four predicted groups.

3. Experiment

3.1. Experimental Procedure

To evaluate whether external knowledge improves performance on NYT Connections puzzles, we evaluated and compared our three models: a baseline Sentence-BERT clustering model, a WordNet-enhanced model, and a WordNet + Wikipedia fusion model. All three models used the same Sentence-BERT encoder and agglomerative clustering framework, allowing us to isolate the effect of adding external knowledge.

Experiments were conducted using the train, validation, and test splits described in Section 2.1. The validation set was used exclusively for hyperparameter tuning and model selection, while the held-out test set was reserved for final evaluation. Each model was given the 16 words from a puzzle and tasked with recovering the four hidden categories through clustering.

Performance was evaluated using Exact Match Accuracy, Pairwise Accuracy, Adjusted Rand Index (ARI), and

Normalized Mutual Information (NMI).

Pairwise Accuracy measures whether the model correctly predicts if every pair of words belongs in the same Connections category. Since each puzzle contains 16 words, this produces 120 pairwise comparisons. The metric provides a partial measure of performance even when a model does not perfectly solve the entire puzzle.

Adjusted Rand Index (ARI) measures how closely the predicted clustering matches the ground-truth Connections groups while correcting for chance agreement. Unlike Exact Match Accuracy, ARI rewards partially correct groupings and is therefore a useful measure of overall clustering quality.

Because NYT Connections is fundamentally a clustering task, ARI was used as the primary metric when selecting model configurations.

3.2. Hyperparameter Tuning

We performed several experiments to identify the strongest configuration for each model. For both the baseline and WordNet-enhanced models, we evaluated three agglomerative clustering linkage methods as a hyperparameter: single, average, and complete linkage. In agglomerative clustering, linkage determines how the distance between two clusters is computed when deciding which clusters to merge.

Single linkage measures the distance between the two closest points in different clusters. While this approach can capture loose semantic relationships, it is prone to a chaining effect in which clusters become connected through a sequence of weak pairwise similarities.

Average linkage computes the average distance between all pairs of points across two clusters. This produces more balanced clusters by considering the overall similarity structure rather than individual word pairs.

Complete linkage measures the distance between the two most distant points in different clusters. As a result, all words within a cluster must be relatively similar to one another before clusters are merged, leading to tighter and more compact groupings.

Because NYT Connections categories typically consist of four strongly related words, complete linkage aligns well with the structure of the task by discouraging clusters that contain only partially related terms. Validation experiments showed that complete linkage achieved the highest Adjusted Rand Index (ARI), and it was therefore selected for final evaluation.

For the WordNet + Wikipedia (Multi-Enhanced) fusion model, we explored a larger hyperparameter space. First, we evaluated different Wikipedia summary lengths by truncating retrieved summaries to 100, 200, and 300 characters. We then performed a grid search over fusion weights controlling the relative contributions of the raw word embedding, WordNet embedding, and Wikipedia embedding.

Validation experiments showed that 200-character Wikipedia summaries consistently outperformed both shorter and longer alternatives. The strongest overall configuration used complete linkage with fusion weights of

$$\alpha = 0.35, \quad \beta = 0.40, \quad \gamma = 0.25$$

and was therefore selected for final evaluation on the test set. Figure 4 summarizes the highest-performing hyperparameter configurations evaluated during model selection.

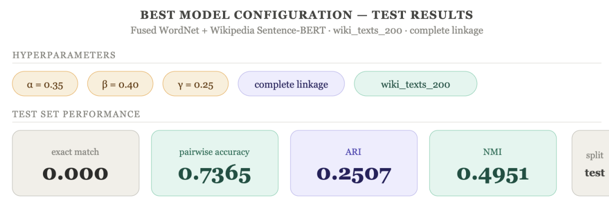

Figure 4. Optimal Multi-Enhanced model configuration showing hyperparameter tuning results: complete linkage, a 200-character Wikipedia window, and an empirical weight distribution of 35% raw word, 40% WordNet, and 25% Wikipedia text context.

Here, α determines the contribution of the original word embedding, β determines the contribution of the WordNet-enhanced embedding, and γ determines the contribution of the Wikipedia-enhanced embedding. By varying these weights, we can control the balance between pretrained semantic knowledge and externally retrieved contextual information.

3.3. Model Comparison

After selecting the best configuration for each model, we evaluated all three approaches on the held-out test set. The baseline Sentence-BERT model achieved an ARI of 0.198 and a pairwise accuracy of 0.710. Augmenting words with WordNet definitions improved performance across all metrics, increasing ARI to 0.230 and pairwise accuracy to 0.729. The strongest results were achieved by the WordNet + Wikipedia fusion model, which reached an ARI of 0.251, NMI of 0.495, and pairwise accuracy of 0.737.

Figures 5 presents the final model comparison results and statistical significance analysis.

4. Results

The highest-performing model was the Multi-Enhanced Sentence-BERT architecture, which combined raw word embeddings with contextual information from both WordNet and Wikipedia. Hyperparameter tuning revealed that the optimal configuration assigned 35% of the final embedding weight to the original word embedding, while the remaining 65% was derived from external knowledge sources. [Figure 4] Specifically, 40% of the total weight

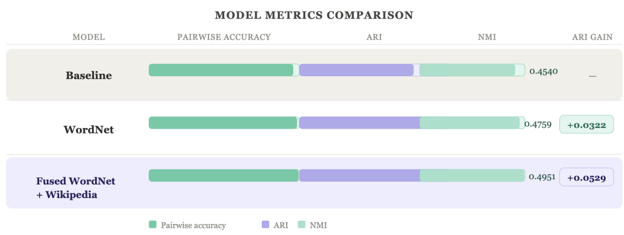

Figure 5. Model performance metrics comparison on the held-out evaluation dataset. The Multi-Enhanced approach utilizing fused lexical and encyclopedic context achieves a 26.8% relative improvement in ARI over the un-augmented baseline.

was assigned to WordNet-enhanced embeddings and 25% to Wikipedia-enhanced embeddings.

These results suggest that contextual information contributed more to successful puzzle solving than the raw word representation itself. Furthermore, the strongest Wikipedia-enhanced model used summaries truncated to 200 characters rather than 300 characters. This finding indicates that concise contextual information is often more useful than longer descriptions, which may introduce irrelevant details and increase noise within the embedding space. Overall, the optimal hyperparameter configuration demonstrates that external lexical and encyclopedic knowledge plays a substantial role in accurately identifying Connections categories.

4.1. Performance Improvements

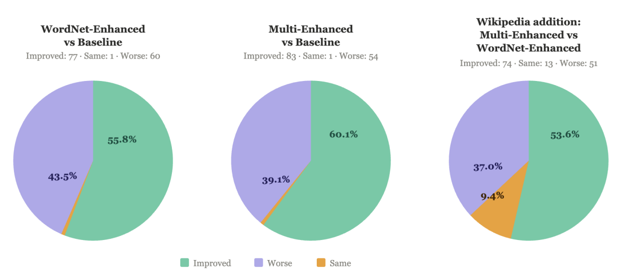

Figure 6. Distribution of puzzle-level performance shifts across experimental conditions. Green segments denote puzzles where clustering quality improved, purple segments indicate performance degradation due to localized context noise, and orange segments represent unchanged outcomes.

Our results demonstrate that external knowledge can substantially improve performance on semantic grouping tasks such as NYT Connections. While the baseline Sentence-BERT model achieved an average ARI of 0.198 and pairwise accuracy of 0.710, the best-performing multi-enhanced model increased ARI to 0.251 and pairwise accuracy to 0.737. This corresponds to a 26.8% relative improvement in ARI, indicating that augmenting embeddings with lexical and encyclopedic context allows the model to recover relationships that are not captured by semantic similarity alone.

As shown in Figure 6, the average improvement masks substantial differences in accuracy across puzzles. Context augmentation (for the multi-enhanced model) improved performance on roughly 60% of puzzles compared to our baseline model, while degrading performance on the remainder, suggesting that external knowledge is highly beneficial when relevant but can introduce noise when retrieved information does not align with the intended category structure.

4.2. Statistical Significance

The statistical significance analysis further supports this conclusion. The WordNet-only model produced only a modest improvement over the baseline and did not achieve statistical significance ($p = 0.090$). This suggests that lexical definitions alone are insufficient to consistently improve clustering performance across all puzzle types. While WordNet provides useful information for ambiguous words, it can also introduce irrelevant senses that increase noise in the embedding space.

In contrast, the fused WordNet + Wikipedia model significantly outperformed both the baseline model ($p = 0.0004$) and the WordNet-only model ($p = 0.033$). These results and their statistical significance indicate that encyclopedic knowledge provides information that is complementary to lexical definitions. Many Connections categories rely on cultural, historical, geographic, or entity-specific knowledge that is not present in standard word embeddings and is only partially represented by dictionary definitions. Wikipedia summaries provide this missing context, enabling the model to better identify relationships among people, places, organizations, brands, and other culturally dependent concepts.

5. Conclusion

We investigated whether external knowledge improves a model's ability to solve NYT Connections puzzles by comparing a baseline Sentence-BERT clustering model against WordNet-enhanced and WordNet + Wikipedia-enhanced approaches. Our results demonstrate that augmenting semantic embeddings with external knowledge improves a model's ability to recover the hidden relationships underlying Connections categories.

More broadly, our findings suggest that many failures on NYT Connections stem not from weak semantic representations alone, but from missing contextual knowledge. While pretrained embeddings capture general semantic similarity, they often lack the lexical and encyclopedic information needed to identify culturally dependent categories, named entities, and ambiguous terms. By incorporating external knowledge sources, our models were able to better capture these relationships and produce more accurate groupings.

Overall, this work highlights the value of knowledge augmentation for semantic grouping tasks. Rather than relying solely on information encoded within pretrained embeddings, models can benefit from lightweight external knowledge sources that provide additional context. These findings suggest that combining semantic representations with relevant external knowledge is a promising direction for improving performance on tasks that require both language understanding and world knowledge.

5.1. Limitations

Several limitations should be considered when interpreting these results. First, only WordNet and Wikipedia were explored as external knowledge sources, and other resources may provide richer or more targeted contextual information. Second, all experiments relied on fixed pretrained Sentence-BERT embeddings; the embedding model itself was not fine-tuned on Connections-style tasks. Third, agglomerative clustering was used throughout the study to maintain consistency across experiments, but alternative clustering approaches may yield stronger performance. Finally, the contextual augmentation process was not selective and frequently incorporated multiple senses of ambiguous words, which occasionally introduced irrelevant information and reduced clustering quality.

5.2. Learnings

Beyond the performance improvements observed in the final models, this project provided several important insights into the process of evaluating knowledge augmentation systems. Initially, our experimental design compared only a baseline Sentence-BERT model against a combined WordNet + Wikipedia model. However, as development progressed, we recognized that this comparison made it difficult to determine which knowledge source was responsible for any observed improvements. To address this limitation, we introduced an intermediate WordNet-only model, allowing us to isolate the contribution of lexical knowledge before evaluating the combined effects of lexical and encyclopedic information.

This additional model proved valuable for interpreting the results. While the WordNet-only model produced only modest improvements over the baseline, the combined WordNet + Wikipedia model achieved substantially stronger performance. This progression revealed that external knowledge is not uniformly useful and that different sources contribute different types of information. WordNet primarily provides lexical and definitional context, whereas Wikipedia contributes broader world knowledge and cultural information that frequently appears in Connections puzzles.

Another key lesson was the importance of hyperparameter analysis. The strongest-performing configuration was not the one that incorporated the largest amount of external context. Instead, performance improved when context sources were carefully weighted and when Wikipedia summaries were truncated to shorter lengths. This demonstrated that adding more information does not necessarily improve model performance and that relevant context is often more valuable than larger quantities of context. More broadly, this project highlighted the importance of understanding how individual system components contribute to performance rather than evaluating only end-to-end results.

5.3. Future Work

Several promising directions remain for future work. One avenue would be to incorporate retrieval-augmented generation (RAG) techniques, where contextual information is dynamically retrieved based on the relationships among all puzzle words rather than independently for each word. Such an approach could allow the model to retrieve more targeted knowledge and reduce the noise introduced by irrelevant definitions or summaries.

Another direction would be to incorporate gameplay-inspired reasoning strategies. Human players of NYT Connections often use partial solutions and category hints to progressively refine their groupings. Future systems could simulate this process by identifying highly confident word groups first and then using the remaining words and inferred category information to guide subsequent clustering decisions. Incorporating the puzzle's currently available "one-away" feedback mechanism could be particularly valuable, as it provides information about groups that are nearly correct and could help the model iteratively improve exact-match performance.

Future research could also explore more advanced retrieval sources such as knowledge graphs, Wikidata, ConceptNet, or large language models capable of generating context on demand. Additionally, fine-tuning embedding models directly on Connections-style tasks, exploring graph-based clustering approaches, or integrating reasoning-oriented language models may further improve performance. Ultimately, the results of this study suggest that semantic grouping tasks benefit from both strong semantic representations and access to relevant external knowledge, making knowledge-augmented reasoning a promising area for continued investigation.

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